	IVW (fixed)	IVW (random)	MR-Egger	Weighted median	PRESSO outlier-removed	MRlap corrected	MVMR + height	Sign concordance
BMD total → fracture (control)	● **-0.337**	● **-0.337**	● **-0.337**	◊ -0.393	● **-0.352**	● **-0.141**	● **-0.343**	**7/7**
Femoral-neck BMD → fracture (control)	● **-0.351**	● **-0.351**	● **-0.343**	◊ -0.365	● **-0.345**	● **-0.123**	● **-0.361**	**7/7**
BMD total → knee OA	● **+0.078**	● **+0.078**	● **+0.078**	◊ +0.065	● **+0.088**	● **+0.028**	● **+0.063**	**7/7**
Lumbar-spine BMD → knee OA	● **+0.095**	● **+0.095**	● **+0.089**	◊ +0.115	● **+0.113**	● **+0.041**	● **+0.068**	**7/7**
Arms BMD → knee OA	● **+0.075**	○ +0.075	○ +0.075	◊ +0.026	● **+0.087**	● **+0.034**	● **+0.065**	**7/7**
Legs BMD → knee OA	● **+0.060**	○ +0.060	○ +0.058	◊ +0.026	● **+0.049**	● **+0.030**	● **+0.040**	**7/7**
Femoral-neck BMD → knee OA	● **+0.058**	○ +0.058	○ +0.058	◊ +0.052	● **+0.071**	● **+0.027**	● **+0.049**	**7/7**
Knee OA → fracture	○ +0.027	○ +0.027	○ +0.031	◊ **+0.019**	○ +0.059	○ +0.040	n/a	**6/6**
Knee OA → BMD total	○ +0.031	○ +0.031	○ +0.036	◊ **+0.026**	○ +0.051	● **+0.344**	n/a	**6/6**
Knee OA → BMD femoral neck	○ +0.042	○ +0.042	○ +0.048	◊ **+0.023**	○ +0.042	● **+0.321**	n/a	**6/6**

● p<0.05 ○ p≥0.05 ◊ direction only (no SE in source output; significance not assessed) n/a estimator not applicable to the pair

Cell shading shows the sign of the estimate (warm = positive, cool = negative). IVW fixed/random, MR-Egger, weighted median: Table 3. PRESSO outlier-removed: IVW recomputed on harmonised SNP data after removing MR-PRESSO outliers (0–3 per pair). MRlap: overlap-corrected estimate. MVMR: additionally conditioned on height (n_h=100/EUR).